EcoLoop Presentation Script

Use this document as your speaker notes during the demonstration. Each page is one slide with main points and talking prompts.

Slide 1: Introduction

- Project name: EcoLoop
- Purpose: a smart waste management dashboard with incentives
- Roles supported: Administrator, Collector, Resident

Notes: Briefly introduce the team and the system goal: structured waste collection, better coordination, and rewards for users.

Slide 2: Problem Statement

- Waste segregation is poorly tracked in many communities
- Collectors and residents lack a shared operational dashboard
- Incentives are not linked to actual waste behavior

Notes: Describe the real-world problem and why this project is needed.

Slide 3: Objectives

- Create a role-based system for waste reporting and collection
- Make data visible through dashboards and analytics
- Provide incentives for users who practice proper segregation

Notes: Explain the general and specific goals of EcoLoop.

Slide 4: Core Features

- Secure login for residents, collectors, and administrators
- Waste submission and pickup scheduling workflows
- Points, rewards, notifications, and reporting

Notes: Highlight the primary functionality available in the application.

Slide 5: Role-Based Features

- Administrator: user management, analytics, notifications
- Collector: route assignments, pickup status updates
- Resident: submit waste entries, view points and rewards

Notes: Show how each user type gets a tailored dashboard experience.

Slide 6: System Architecture

- Angular frontend handles the user interface and navigation
- Node.js + Express backend processes API requests
- MySQL database stores users, submissions, pickups, and rewards

Notes: Explain the flow: frontend !' backend !' database.

Slide 7: ACID and Data Integrity

- Transactions ensure atomic operations for critical changes
- Consistency checks validate data before each update
- Isolation and locks prevent concurrent write conflicts
- Durability is supported through backups and DB persistence

Notes: Mention that the backend is designed for reliable data handling.

Slide 8: Demo Plan

- Open the login page and enter credentials
- Show the administrator and resident dashboard views
- Demonstrate submitting a waste entry and checking analytics

Notes: Keep the demo focused, step-by-step, and use real screens.

Slide 9: Tools & Technologies

- Frontend: Angular, TypeScript, HTML, CSS
- Backend: Node.js, Express, mysql2, MySQL
- Development environment: VS Code, Git, npm

Notes: List the technologies used to build the project.

Slide 10: Challenges Faced

- Startup error handling and dependency alignment
- Ensuring backend MySQL connection and database schema
- Creating a polished, responsive interface for multiple roles

Notes: Talk briefly about the main development challenges and how they were solved.

Slide 11: Results

- Functional waste management dashboard with role-specific views
- Live data operations including submissions, pickups, and rewards
- Working backend with MySQL integration and ACID support

Notes: Summarize what was accomplished in the project.

Slide 12: Future Enhancements

- Add map-based pickup routing and live tracking
- Implement mobile-first design for field workers
- Add reporting exports and advanced analytics

Notes: Close the presentation with possible next steps for EcoLoop.

Thank you for listening. We are ready to answer questions.